# P8131 Homework 3

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## Problem 1: Esophageal Cancer and Alcohol Consumption

**The following table gives data from a retrospective study between esophageal cancer and daily alcohol consumption adjusted for age.**

| Age | Case 0-79g | Case 80+g | Control 0-79g | Control 80+g |
|---|---|---|---|---|
| 25-34 | 0 | 1 | 106 | 9 |
| 35-44 | 5 | 4 | 164 | 26 |
| 45-54 | 21 | 25 | 138 | 29 |
| 55-64 | 34 | 42 | 139 | 27 |
| 65-74 | 36 | 19 | 88 | 18 |
| 75+ | 8 | 5 | 31 | 0 |

```r
# Input data
# Create data frame with all cells
age_midpoint <- c(25, 35, 45, 55, 65, 75)

# Case data
case_low <- c(0, 5, 21, 34, 36, 8)   # 0-79g alcohol
case_high <- c(1, 4, 25, 42, 19, 5)  # 80+g alcohol

# Control data
ctrl_low <- c(106, 164, 138, 139, 88, 31)   # 0-79g alcohol
ctrl_high <- c(9, 26, 29, 27, 18, 0)        # 80+g alcohol

# Create long-format data for prospective analysis
# Each row: age, alcohol (0=low, 1=high), case count, control count
data1 <- data.frame(
  age = rep(age_midpoint, each = 2),
  alcohol = rep(c(0, 1), 6),
  case = c(rbind(case_low, case_high)),
  control = c(rbind(ctrl_low, ctrl_high))
)

data1$total <- data1$case + data1$control
```

**(a) Fit a prospective model to the data to study the relation between alcohol consumption, age, and disease (model age as a continuous variable taking values 25, 35, 45, 55, 65, and 75). Interpret the result.**

For a retrospective (case-control) study, although we sample based on disease status, we can still fit a prospective logistic model to estimate the odds ratio, since the odds ratio is symmetric (same whether estimated prospectively or retrospectively).

The model:

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \cdot \text{Age} + \beta_2 \cdot \text{Alcohol}$$

where $\pi = P(\text{Disease} = 1|\text{Age}, \text{Alcohol})$ and Alcohol $= 1$ for 80+ g/day, 0 otherwise.

```r
# Fit prospective logistic regression model
fit_prospective <- glm(cbind(case, control) ~ age + alcohol,
                       family = binomial(link = "logit"), data = data1)
summary(fit_prospective)
```

```
##
## Call:
## glm(formula = cbind(case, control) ~ age + alcohol, family = binomial(link = "logit"),
##     data = data1)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -5.023449   0.418224 -12.011   <2e-16 ***
## age          0.061579   0.007291   8.446   <2e-16 ***
## alcohol      1.780000   0.187086   9.514   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 211.608  on 11  degrees of freedom
## Residual deviance:  31.932  on  9  degrees of freedom
## AIC: 78.259
##
## Number of Fisher Scoring iterations: 4
```

```r
# Extract coefficients and confidence intervals
coefs <- summary(fit_prospective)$coefficients
ci <- confint.default(fit_prospective)

# Odds ratios
OR_age <- exp(coef(fit_prospective)["age"])
OR_alcohol <- exp(coef(fit_prospective)["alcohol"])

# OR confidence intervals
OR_ci_age <- exp(ci["age", ])
OR_ci_alcohol <- exp(ci["alcohol", ])

cat("Odds Ratio for Age (per 10-year increase):", round(OR_age^10, 4), "\n")
```

```
## Odds Ratio for Age (per 10-year increase): 1.8511
```

```r
cat("95% CI:", round(OR_ci_age^10, 4), "\n\n")
```

```
## 95% CI: 1.6046 2.1355
```

```r
cat("Odds Ratio for Alcohol (80+g vs 0-79g):", round(OR_alcohol, 4), "\n")
```

```
## Odds Ratio for Alcohol (80+g vs 0-79g): 5.9299
```

```r
cat("95% CI:", round(OR_ci_alcohol, 4), "\n")
```

```
## 95% CI: 4.1096 8.5564
```

**Interpretation:**

1. **Age effect:** $\hat{\beta}_1 = 0.06158$ ($p < 0.001$). For each 10-year increase in age, the odds of esophageal cancer increase by a factor of $e^{10\times 0.06158} = 1.851$ (95% CI: 1.605 – 2.135), holding alcohol consumption constant.
2. **Alcohol effect:** $\hat{\beta}_2 = 1.78$ ($p < 0.001$). The odds of esophageal cancer for high alcohol consumption (80+ g/day) are $e^{1.78} = 5.93$ times the odds for low alcohol consumption (0-79 g/day), adjusting for age (95% CI: 4.11 – 8.556).
3. **Model fit:** The residual deviance is 31.93 on 9 degrees of freedom. Comparing to $\chi^2_{0.95}(9) = 16.92$, the deviance exceeds the threshold, suggesting some lack of fit ($p = 2 \times 10^{-4}$). This could be due to the assumption that age has a linear effect on the log-odds, or potential interaction between age and alcohol.

**Conclusion:** Both age and high alcohol consumption are significantly associated with increased risk of esophageal cancer.

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**(b) Let $\Psi_j$ be the odds ratio relating alcohol consumption and disease in the $j^{th}$ age group ($j = 1, ..., 6$). Assume different age groups have different odds. Compare the following two models:**

- $M_0 : \Psi_j = 1$ for all $j$
- $M_1 : \Psi_j = \Psi$ (where $\Psi$ is an unknown constant)

**Model Formulation:**

For each age group $j$, let $\pi_{j,k}$ be the probability of disease given alcohol level $k$ (0 = low, 1 = high). The odds ratio in age group $j$ is:

$$\Psi_j = \frac{\pi_{j,1}/(1-\pi_{j,1})}{\pi_{j,0}/(1-\pi_{j,0})}$$

**Model $M_0$:** $\Psi_j = 1$ for all $j$ means alcohol has no effect on disease within any age group.

$$\text{logit}(\pi) = \beta_0 + \beta_1 \cdot \text{Age}$$

This model includes only age (as a continuous variable), with no alcohol effect.

**Model $M_1$:** $\Psi_j = \Psi$ (constant) means the odds ratio for alcohol is the same across all age groups. For any age group $j$,

$$\text{logit}(\pi_{j,1}) - \text{logit}(\pi_{j,0}) = log(\Psi) = \beta_2$$

which means $\log(\Psi_j) = \beta_2$ is the same for all $j$. Therefore $\Psi_j = e^{\beta_2} = \Psi$ is constant across all age groups, consistent with the logistic regression model structure.

$$\text{logit}(\pi) = \beta_0 + \beta_1 \cdot \text{Age} + \beta_2 \cdot \text{Alcohol}$$

$M_0$ is nested within $M_1$ (set $\beta_2 = 0$ in $M_1$ to get $M_0$).

```r
# Fit M0: Only age effect (no alcohol effect, implies Psi_j = 1)
fit_M0 <- glm(cbind(case, control) ~ age,
              family = binomial(link = "logit"), data = data1)

# Fit M1: Age effect + constant alcohol effect (Psi_j = Psi for all j)
fit_M1 <- glm(cbind(case, control) ~ age + alcohol,
              family = binomial(link = "logit"), data = data1)

# Deviance analysis
dev_M0 <- fit_M0$deviance
dev_M1 <- fit_M1$deviance
df_M0 <- fit_M0$df.residual
df_M1 <- fit_M1$df.residual

# Test statistic: difference in deviances
dev_diff <- dev_M0 - dev_M1
df_diff <- df_M0 - df_M1
p_value <- 1 - pchisq(dev_diff, df_diff)

cat("Model M0 (Psi_j = 1): Deviance =", round(dev_M0, 4), ", df =", df_M0, "\n")
```

```
## Model M0 (Psi_j = 1): Deviance = 124.3149 , df = 10
```

```r
cat("Model M1 (Psi_j = Psi): Deviance =", round(dev_M1, 4), ", df =", df_M1, "\n")
```

```
## Model M1 (Psi_j = Psi): Deviance = 31.9315 , df = 9
```

```r
cat("\nDeviance difference:", round(dev_diff, 4), "\n")
```

```
##
## Deviance difference: 92.3834
```

```r
cat("df difference:", df_diff, "\n")
```

```
## df difference: 1
```

```r
cat("Critical value chi^2_0.95(1):", round(qchisq(0.95, 1), 4), "\n")
```

```
## Critical value chi^2_0.95(1): 3.8415
```

```r
cat("P-value:", format(p_value, scientific = TRUE, digits = 4), "\n")
```

```
## P-value: 0e+00
```

```r
# Estimate of common odds ratio Psi
Psi_hat <- exp(coef(fit_M1)["alcohol"])
ci_Psi <- exp(confint.default(fit_M1)["alcohol", ])
cat("\nEstimate of common odds ratio Psi:", round(Psi_hat, 4), "\n")
```

```
##
## Estimate of common odds ratio Psi: 5.9299
```

```r
cat("95% CI for Psi:", round(ci_Psi, 4), "\n")
```

```
## 95% CI for Psi: 4.1096 8.5564
```

**Deviance Analysis:**

Since $M_0$ is nested within $M_1$, we can use the likelihood ratio test:

$$D_{M_0} - D_{M_1} \sim \chi^2(df_{M_0} - df_{M_1}) \text{ under } H_0 : \beta = 0$$

- Test statistic: 92.3834
- Degrees of freedom: 1
- Critical value $\chi^2_{0.95}(1)$: 3.8415
- P-value: $0e + 00$

**Conclusion:** Since the test statistic (92.38) far exceeds the critical value (3.84), we reject $M_0$ in favor of $M_1$ at the 0.05 significance level. There is strong evidence that alcohol consumption is associated with esophageal cancer risk ($\Psi \neq 1$).

The estimated common odds ratio is $\hat{\Psi} = 5.93$ (95% CI: 4.11 – 8.556), indicating that high alcohol consumption (80+ g/day) is associated with approximately 5.9 times the odds of esophageal cancer compared to low consumption, across all age groups.

# Problem 2: Orobanche Seed Germination

**The following table provides data from a study of the germination of two species of Orobanche seeds. The seeds were grown on 1/125 dilutions of two different root extract media (cucumber or bean) in a 2 x 2 factorial layout with replicates. The data ($y_i/m_i$) consist of the number of seeds, $m_i$, and the number germinating, $y_i$, for each batch. Interest focuses on the possible differences in germination rates for the two types of seed and root extract.**

```r
# Input germination data
# O. aegyptiaca 75
bean_75 <- data.frame(
  germ = c(10, 23, 23, 26, 17),
  total = c(39, 62, 81, 51, 39),
  seed = "O. aegyptiaca 75",
  extract = "Bean"
)

cucumber_75 <- data.frame(
  germ = c(5, 53, 55, 32, 46, 10),
  total = c(6, 74, 72, 51, 79, 13),
```

```r
  seed = "O. aegyptiaca 75",
  extract = "Cucumber"
)

# O. aegyptiaca 73
bean_73 <- data.frame(
  germ = c(8, 10, 8, 23, 0),
  total = c(16, 30, 28, 45, 4),
  seed = "O. aegyptiaca 73",
  extract = "Bean"
)

cucumber_73 <- data.frame(
  germ = c(3, 22, 15, 32, 3),
  total = c(12, 41, 30, 51, 7),
  seed = "O. aegyptiaca 73",
  extract = "Cucumber"
)

# Combine all data
germ_data <- rbind(bean_75, cucumber_75, bean_73, cucumber_73)
germ_data$not_germ <- germ_data$total - germ_data$germ
germ_data$seed <- factor(germ_data$seed)
germ_data$extract <- factor(germ_data$extract)

# Display data summary
kable(germ_data[, c("seed", "extract", "germ", "total")],
      col.names = c("Seed Type", "Extract", "Germinated", "Total"),
      caption = "Orobanche Seed Germination Data")
```

Table 1: Orobanche Seed Germination Data

| Seed Type | Extract | Germinated | Total |
|---|---|---:|---:|
| O. aegyptiaca 75 | Bean | 10 | 39 |
| O. aegyptiaca 75 | Bean | 23 | 62 |
| O. aegyptiaca 75 | Bean | 23 | 81 |
| O. aegyptiaca 75 | Bean | 26 | 51 |
| O. aegyptiaca 75 | Bean | 17 | 39 |
| O. aegyptiaca 75 | Cucumber | 5 | 6 |
| O. aegyptiaca 75 | Cucumber | 53 | 74 |
| O. aegyptiaca 75 | Cucumber | 55 | 72 |
| O. aegyptiaca 75 | Cucumber | 32 | 51 |
| O. aegyptiaca 75 | Cucumber | 46 | 79 |
| O. aegyptiaca 75 | Cucumber | 10 | 13 |
| O. aegyptiaca 73 | Bean | 8 | 16 |
| O. aegyptiaca 73 | Bean | 10 | 30 |
| O. aegyptiaca 73 | Bean | 8 | 28 |
| O. aegyptiaca 73 | Bean | 23 | 45 |
| O. aegyptiaca 73 | Bean | 0 | 4 |
| O. aegyptiaca 73 | Cucumber | 3 | 12 |
| O. aegyptiaca 73 | Cucumber | 22 | 41 |
| O. aegyptiaca 73 | Cucumber | 15 | 30 |

| Seed Type | Extract | Germinated | Total |
|---|---|---|---|
| O. aegyptiaca 73 | Cucumber | 32 | 51 |
| O. aegyptiaca 73 | Cucumber | 3 | 7 |

**(a) Fit a logistic regression model to study the relation between germination rates and different types of seed and root extract. Interpret the result.**

Model:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \cdot \text{Seed} + \beta_2 \cdot \text{Extract}$$

where Seed = 1 for O. aegyptiaca 75, Extract = 1 for Cucumber.

```r
# Fit logistic regression with main effects
fit_germ <- glm(cbind(germ, not_germ) ~ seed + extract,
                family = binomial(link = "logit"), data = germ_data)
summary(fit_germ)
```

```
##
## Call:
## glm(formula = cbind(germ, not_germ) ~ seed + extract, family = binomial(link = "logit"),
##     data = germ_data)
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)            -0.7005     0.1507  -4.648 3.36e-06 ***
## seedO. aegyptiaca 75    0.2705     0.1547   1.748   0.0804 .
## extractCucumber         1.0647     0.1442   7.383 1.55e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 98.719  on 20  degrees of freedom
## Residual deviance: 39.686  on 18  degrees of freedom
## AIC: 122.28
##
## Number of Fisher Scoring iterations: 4
```

```r
# Odds ratios and confidence intervals
OR <- exp(coef(fit_germ))
ci <- exp(confint.default(fit_germ))

results <- data.frame(
  Coefficient = round(coef(fit_germ), 4),
  OR = round(OR, 4),
  CI_lower = round(ci[, 1], 4),
  CI_upper = round(ci[, 2], 4),
  p_value = round(summary(fit_germ)$coefficients[, 4], 4)
)
kable(results, caption = "Logistic Regression Results (Standard Binomial)")
```

Table 2: Logistic Regression Results (Standard Binomial)

| | Coefficient | OR | CI_lower | CI_upper | p_value |
|---|---|---|---|---|---|
| (Intercept) | -0.7005 | 0.4963 | 0.3694 | 0.6669 | 0.0000 |
| seedO. aegyptiaca 75 | 0.2705 | 1.3106 | 0.9678 | 1.7748 | 0.0804 |
| extractCucumber | 1.0647 | 2.9001 | 2.1860 | 3.8474 | 0.0000 |

**Interpretation:**

1. **Seed type effect:** The coefficient for seed type (O. aegyptiaca 75 vs 73) is $\hat{\beta}_1 = 0.2705$ with OR = 1.311. O. aegyptiaca 75 has 1.311 times the odds of germination compared to O. aegyptiaca 73 (p = 0.0804). Note that the 95% CI (0.968 – 1.775) includes 1, suggesting the effect may not be statistically significant.

2. **Extract effect:** The coefficient for extract type (Cucumber vs Bean) is $\hat{\beta}_2 = 1.0647$ with OR = 2.9. Seeds grown on cucumber extract have 2.9 times the odds of germination compared to bean extract (p = 0).

3. **Model deviance:** Residual deviance = 39.69 on 18 df. Compared to $\chi^2_{0.95}(18) = 28.87$, the deviance is much larger, suggesting lack of fit which could indicate overdispersion.

---

**(b) Is there overdispersion? If so, what is the estimate of dispersion parameter? Update your model and reinterpret the result.**

```r
# Check for overdispersion
# Pearson chi-squared statistic
pearson_resid <- residuals(fit_germ, type = "pearson")
G <- sum(pearson_resid^2)
n <- nrow(germ_data)
p <- length(coef(fit_germ))

# Estimate dispersion parameter
phi_hat <- G / (n - p)

cat("Pearson chi-squared statistic G =", round(G, 4), "\n")
```

```
## Pearson chi-squared statistic G = 38.3106
```

```r
cat("Degrees of freedom (n - p) =", n - p, "\n")
```

```
## Degrees of freedom (n - p) = 18
```

```r
cat("Estimated dispersion parameter phi_hat = G/(n-p) =", round(phi_hat, 4), "\n")
```

```
## Estimated dispersion parameter phi_hat = G/(n-p) = 2.1284
```

```r
cat("Alternative estimate using deviance: D/(n-p) =",
    round(fit_germ$deviance / (n - p), 4), "\n")
```

```
## Alternative estimate using deviance: D/(n-p) = 2.2048
```

**Assessment of overdispersion:**

Under the standard binomial model, $E[G] \approx n - p = 18$ when the model is correct. We observe $G = 38.31$, which is substantially larger than expected.

The estimated dispersion parameter is:

$$\hat{\phi} = \frac{G}{n-p} = \frac{38.31}{18} = 2.128$$

Since $\hat{\phi} = 2.128 > 1$, there is evidence of **overdispersion** in the data.

```r
# Adjust for overdispersion by specifying dispersion parameter
summary_adj <- summary(fit_germ, dispersion = phi_hat)
summary_adj
```

```
##
## Call:
## glm(formula = cbind(germ, not_germ) ~ seed + extract, family = binomial(link = "logit"),
##     data = germ_data)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)             -0.7005     0.2199  -3.186  0.00144 **
## seedO. aegyptiaca 75     0.2705     0.2257   1.198  0.23081
## extractCucumber          1.0647     0.2104   5.061 4.18e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 2.128368)
##
##     Null deviance: 98.719  on 20  degrees of freedom
## Residual deviance: 39.686  on 18  degrees of freedom
## AIC: 122.28
##
## Number of Fisher Scoring iterations: 4
```

```r
# Updated results with overdispersion adjustment
# Note: coefficients are the same, only SEs and p-values change
se_adj <- summary_adj$coefficients[, 2]
ci_adj <- cbind(
  coef(fit_germ) - 1.96 * se_adj,
  coef(fit_germ) + 1.96 * se_adj
)

results_adj <- data.frame(
  Coefficient = round(coef(fit_germ), 4),
  SE = round(se_adj, 4),
```

```r
  OR = round(OR, 4),
  CI_lower = round(exp(ci_adj[, 1]), 4),
  CI_upper = round(exp(ci_adj[, 2]), 4),
  p_value = round(summary_adj$coefficients[, 4], 4)
)
kable(results_adj,
      caption = "Results Adjusted for Overdispersion")
```

Table 3: Results Adjusted for Overdispersion

| | Coefficient | SE | OR | CI_lower | CI_upper | p_value |
|---|---|---|---|---|---|---|
| (Intercept) | -0.7005 | 0.2199 | 0.4963 | 0.3226 | 0.7638 | 0.0014 |
| seedO. aegyptiaca 75 | 0.2705 | 0.2257 | 1.3106 | 0.8420 | 2.0397 | 0.2308 |
| extractCucumber | 1.0647 | 0.2104 | 2.9001 | 1.9201 | 4.3803 | 0.0000 |

**Comparison of standard errors:**

```r
# Compare standard errors
se_comparison <- data.frame(
  Parameter = names(coef(fit_germ)),
  SE_binomial = round(summary(fit_germ)$coefficients[, 2], 4),
  SE_adjusted = round(se_adj, 4),
  Ratio = round(se_adj / summary(fit_germ)$coefficients[, 2], 4)
)
kable(se_comparison, caption = "Standard Error Comparison")
```

Table 4: Standard Error Comparison

| | Parameter | SE_binomial | SE_adjusted | Ratio |
|---|---|---|---|---|
| (Intercept) | (Intercept) | 0.1507 | 0.2199 | 1.4589 |
| seedO. aegyptiaca 75 | seedO. aegyptiaca 75 | 0.1547 | 0.2257 | 1.4589 |
| extractCucumber | extractCucumber | 0.1442 | 0.2104 | 1.4589 |

**Reinterpretation with overdispersion adjustment:**

After accounting for overdispersion ($\hat{\phi} = 2.128$):

1. **Seed type effect:** O. aegyptiaca 75 still has higher germination odds than O. aegyptiaca 73 (OR = 1.311), but the effect is not statistically significant at $\alpha = 0.05$ (p = 0.2308).

2. **Extract effect:** Cucumber extract is associated with higher germination odds than bean extract (OR = 2.9), and this effect remains statistically significant at $\alpha = 0.05$ (p = 0).

3. **Standard errors:** The standard errors are inflated by a factor of $\sqrt{\hat{\phi}} = \sqrt{2.128} = 1.459$ compared to the standard binomial model.

**Conclusion:** After adjusting for overdispersion, extract type remains statistically significant while seed type does not. Cucumber extract is associated with higher germination rates than bean extract.

---

**(c) What is a plausible cause of the overdispersion?**

1. **Intra-batch correlation:** Seeds within the same batch share environmental conditions, leading to correlated outcomes ($\mathrm{Var}(Y) = m\pi(1-\pi)\{1+(m-1)\rho\} > m\pi(1-\pi)$).

2. **Batch-to-batch heterogeneity:** Unmeasured factors (seed quality, storage conditions) may cause germination probability to vary across batches beyond what seed type and extract explain.